



Citius Con: An Event for Postgres

FIBONACCI SPIRALS & WAYS TO CONTRIBUTE TO POSTGRES—BEYOND CODE

Claire Giordano

@clairegiordano • @citiusdata • @AzureDBPostgres



Leonardo Pisano Bigollo

(1170–1240)

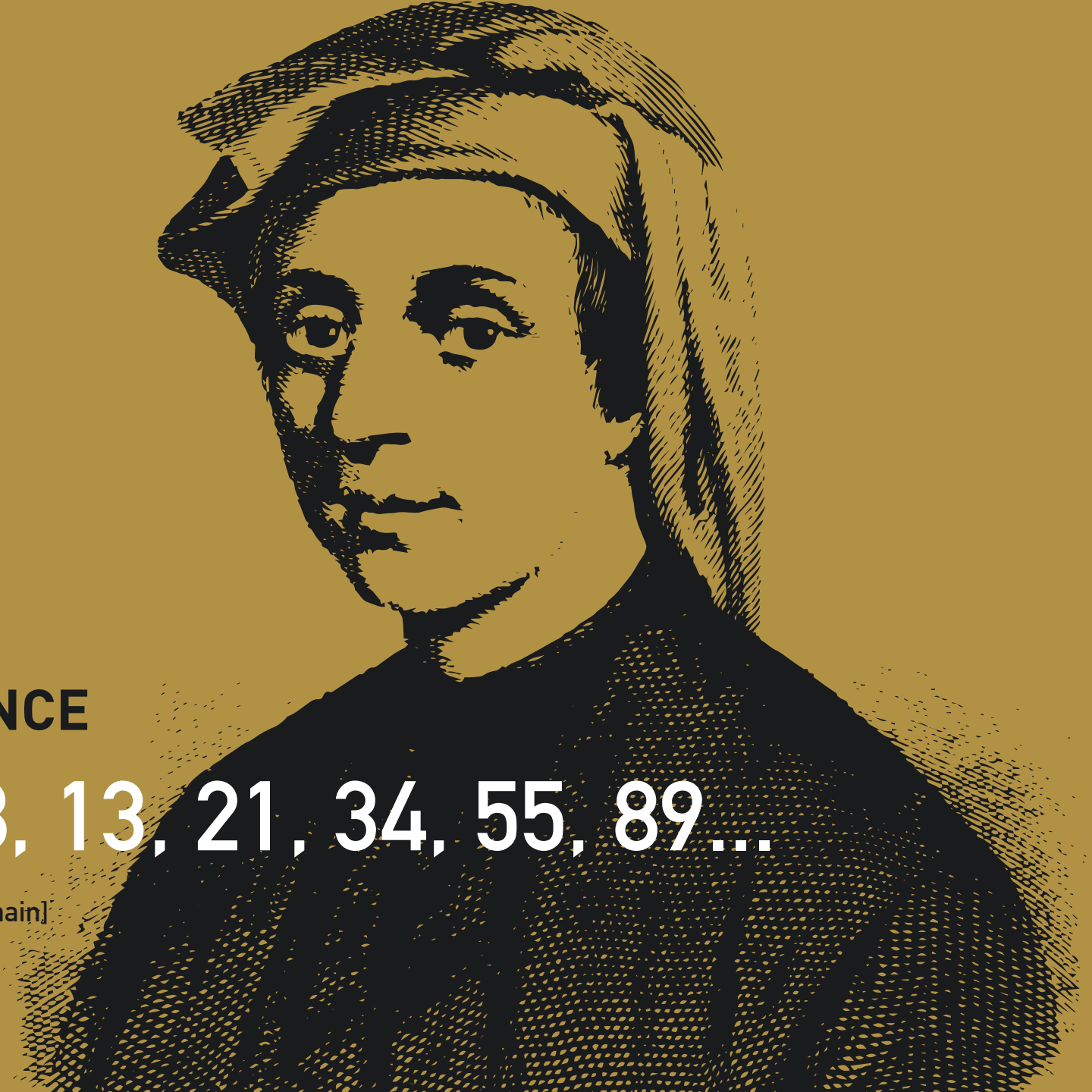
Source: Wikimedia Commons [Public Domain]



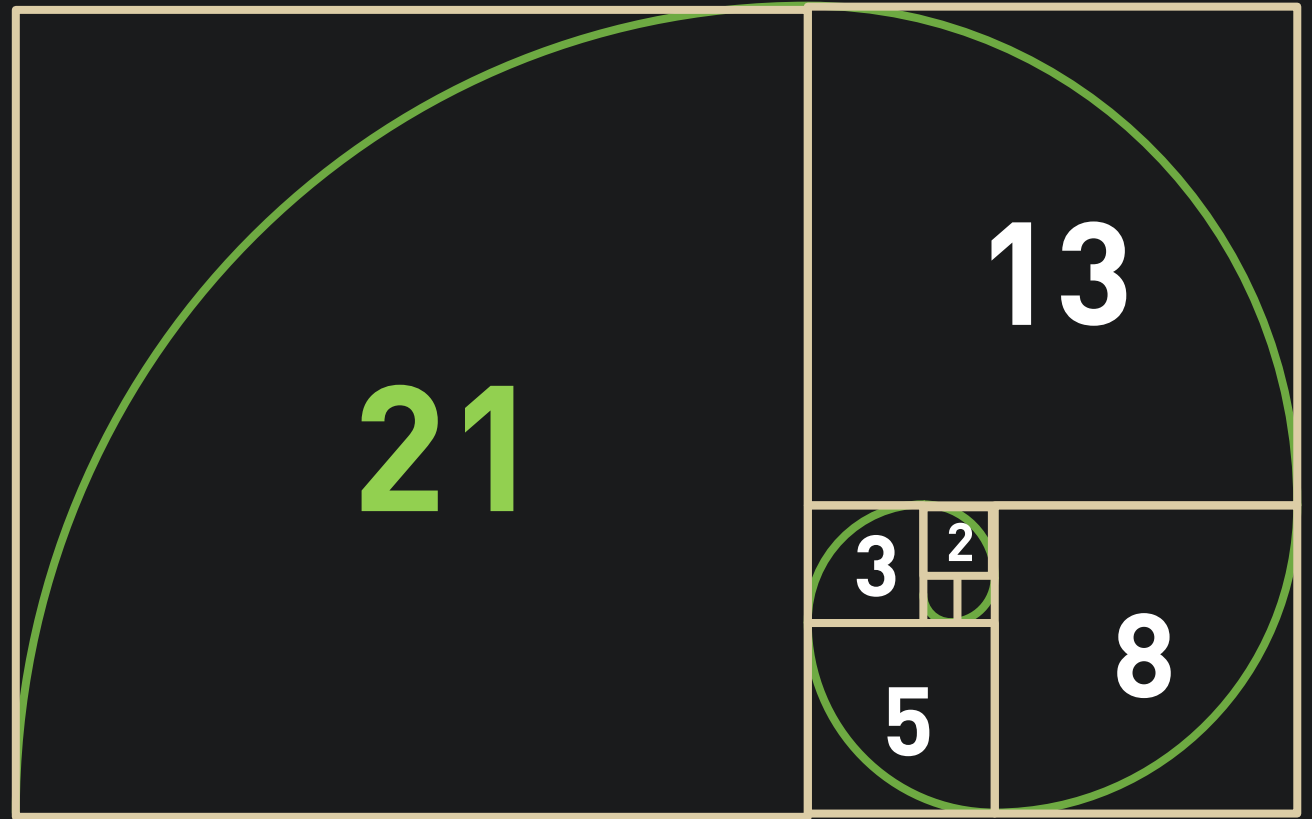
FIBONACCI SEQUENCE

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89...

Source: Wikimedia Commons [Public Domain]



What do Fibonacci
spirals have to do
with contributing to
Postgres?



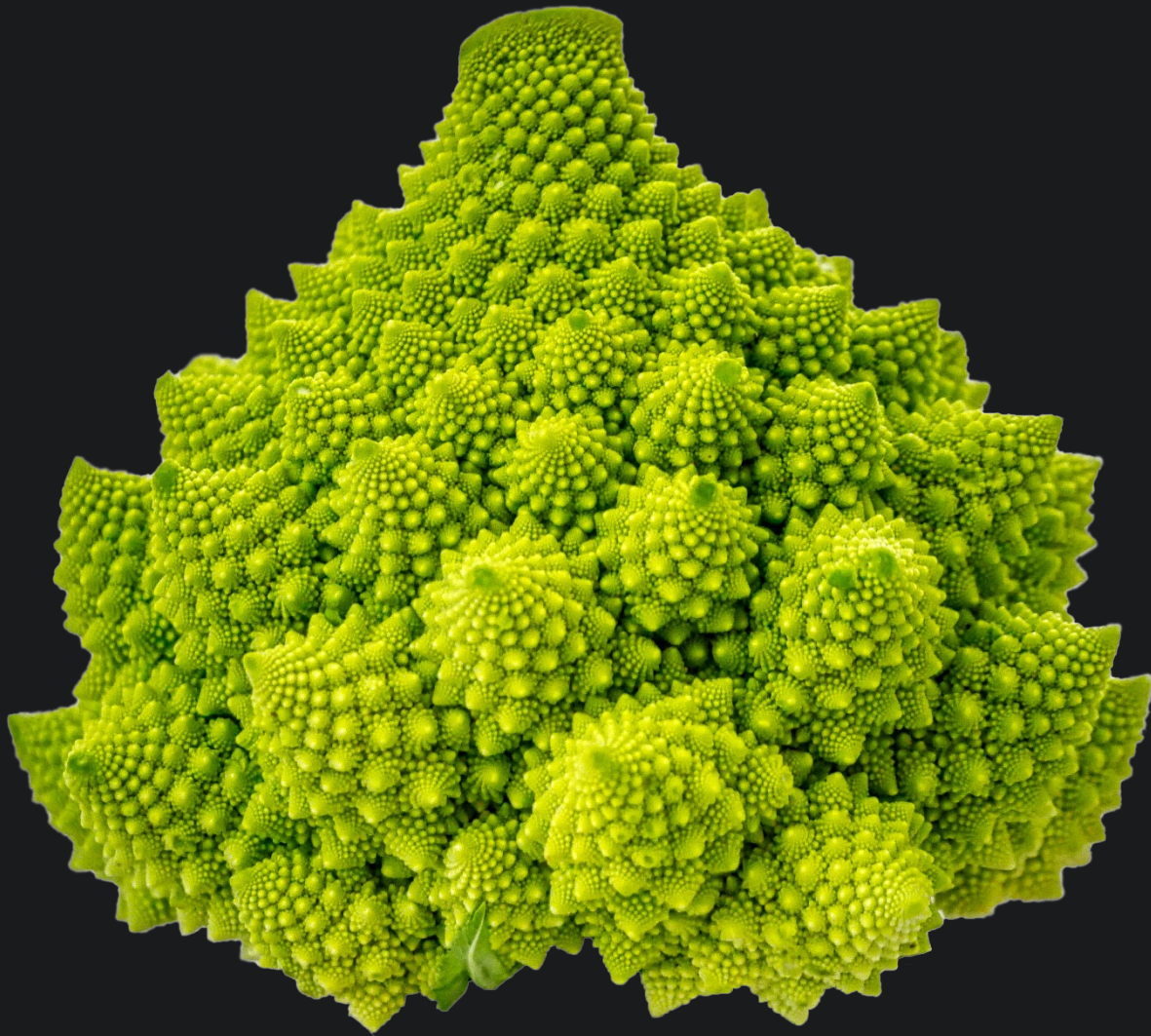
1, 1, 2, 3, 5, 8, 13, 21, 34, ...

BECAUSE I ALWAYS WANTED



TO USE A ROMANESCO IN A TALK

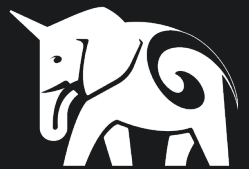
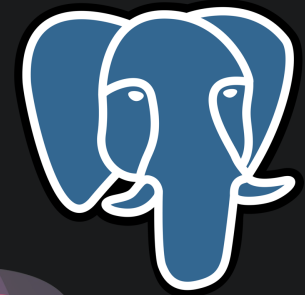
FIBONACCI SPIRAL & FRACTAL



WAYS TO CONTRIBUTE TO POSTGRES— BEYOND CODE

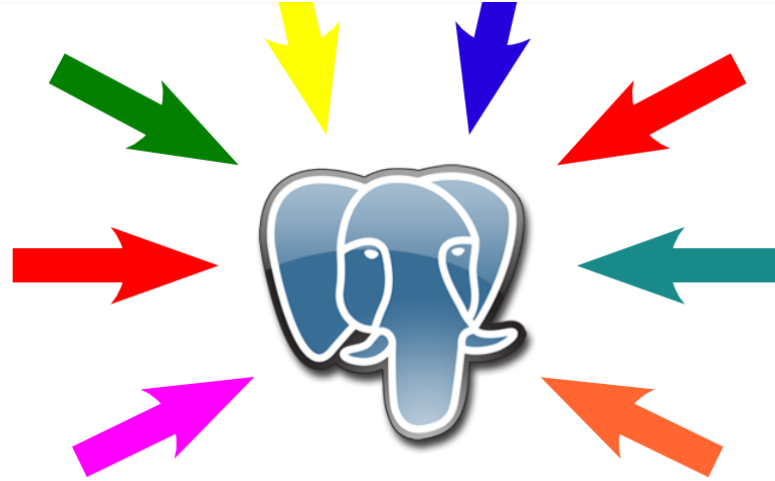


@clairegiordano



Why "Beyond Code"?
What if... you're not a developer?

Josh Berkus gave a similar talk in ~2013

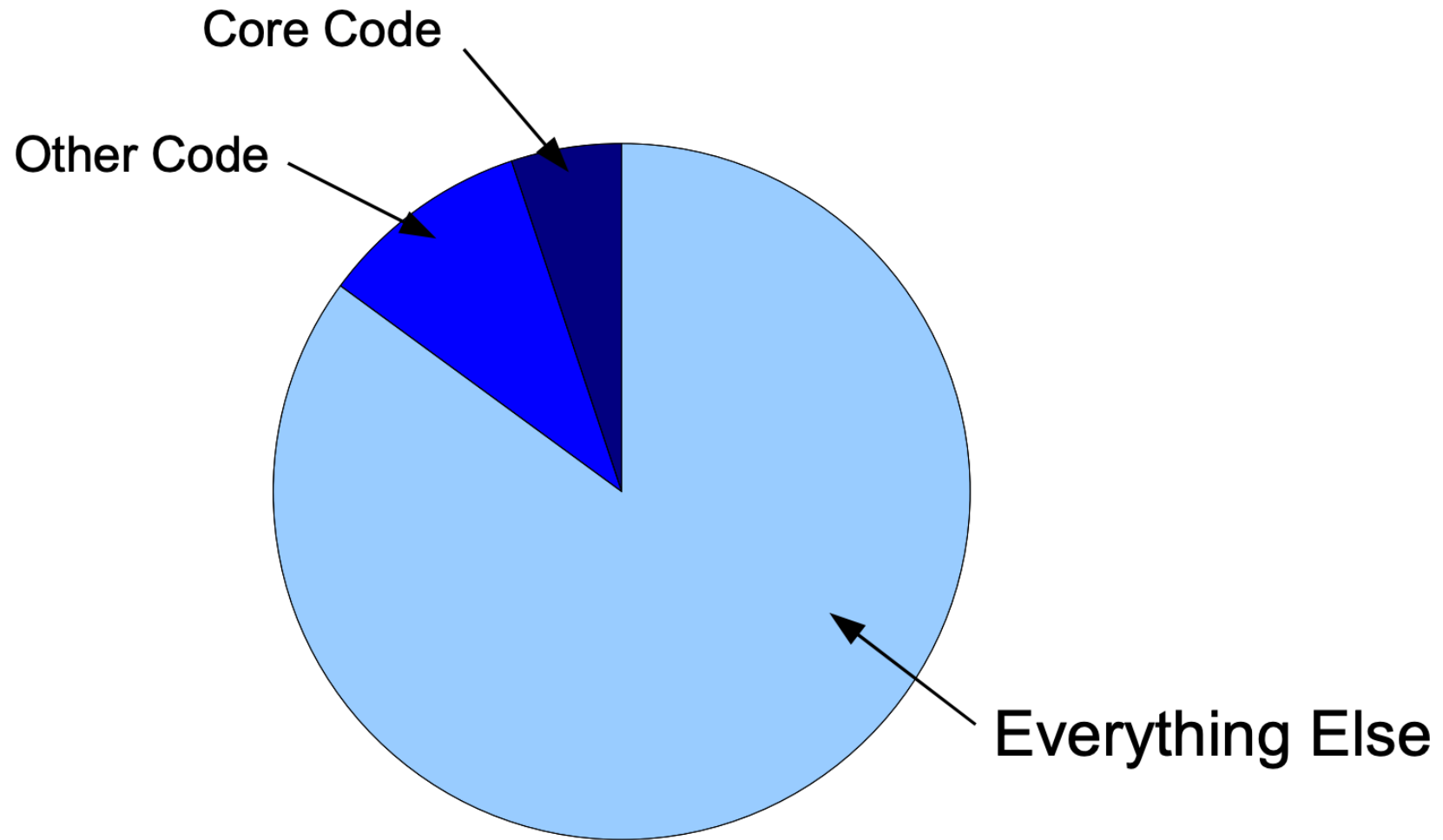


How to contribute to PostgreSQL

or, 50 Ways To Love Your Project

Source: Josh Berkus on GitHub: https://jberkus.github.io/pdf_presos/50_ways.pdf

This is how Josh drew "Contribution Pie" in 2013



Why Beyond Code?

Just like racing sailboats

Every person in the crew—every
role—matters

Source: Livestream of Rolex Big Boats Racing Series, Sep 2021
(my son is on the crew!)



18 Ways to Contribute to Postgres—Beyond Code

1. Write
2. Give a Talk
3. Write a book
4. Video
5. Outreach to other dev communities
6. User stories
7. Promote, tweet too
8. Run & support Meetups
9. Organize a PG event
10. Shine light on others
11. Report bugs
12. User testing of stuff
13. Translations
14. Doc feedback
15. Stickers
16. Design skills
17. #sketchnotes
18. Welcome people in, & answer lots of questions

@clairegiordano

1



Write a blog post

Publish an article

Share learnings & expertise

Shine light on value of Postgres

1

Syndicate to Planet Postgres

- to increase reach of your posts
- @planetpostgres on Twitter, too



The image shows a screenshot of the Planet PostgreSQL Twitter profile. The profile features a blue elephant icon as the profile picture. The name is "Planet PostgreSQL" and the handle is "@planetpostgres". The bio states "Blog entries from members of the PostgreSQL community". The location is "Anywhere!" and the website is "planet.postgresql.org". The account was joined in June 2008. It shows 1 following and 29.3K followers. A list of users followed by the account is visible, including Lotte Feliuss, PostgreSQL Stan, and 164 others. The tabs for Tweets, Tweets & replies, Media, and Likes are at the bottom. The first tweet is partially visible, showing the Planet PostgreSQL profile picture and the text "Hubert 'depez' Lubaczewski: Waiting for".



 **Following**

Planet PostgreSQL
@planetpostgres

Blog entries from members of the PostgreSQL community

 Anywhere!  planet.postgresql.org

 Joined June 2008

1 Following **29.3K Followers**

 Followed by Lotte Feliuss, PostgreSQL Stan, and 164 others you follow

Tweets Tweets & replies Media Likes

 **Planet Postgre...** @planetpostg... · Apr 8 ...
Hubert 'depez' Lubaczewski: Waiting for

Share your expertise—give a talk



A promotional banner for FOSDEM 2020. The background is a blurred image of a large crowd at a conference. In the center is a white gear icon with two dots inside, resembling a face. Text is arranged around and below the gear. At the bottom, there is a dark grey bar with the event location and dates, and a purple button for the schedule.

beer
open source
lightning talks

devrooms
8000+ hackers
835 lectures

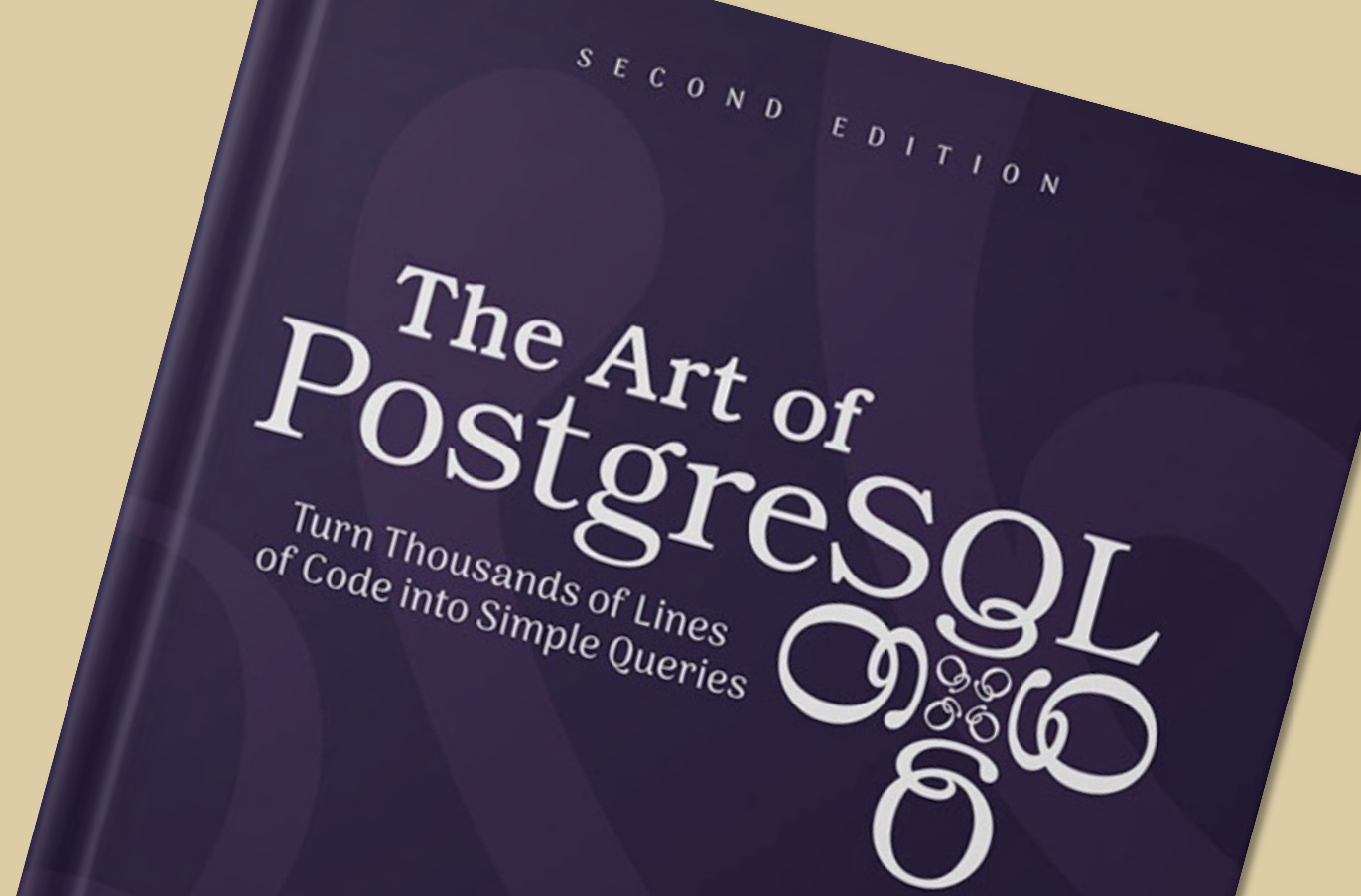
Brussels / 1 & 2 February 2020 [schedule](#)

Source: fosdem.org/2020

3

Write a Book

make it easier for newcomers to use Postgres



CLAIRE15 ~ a 15% off discount on any edition

Source: theartofpostgresql.com

@clairegiordano

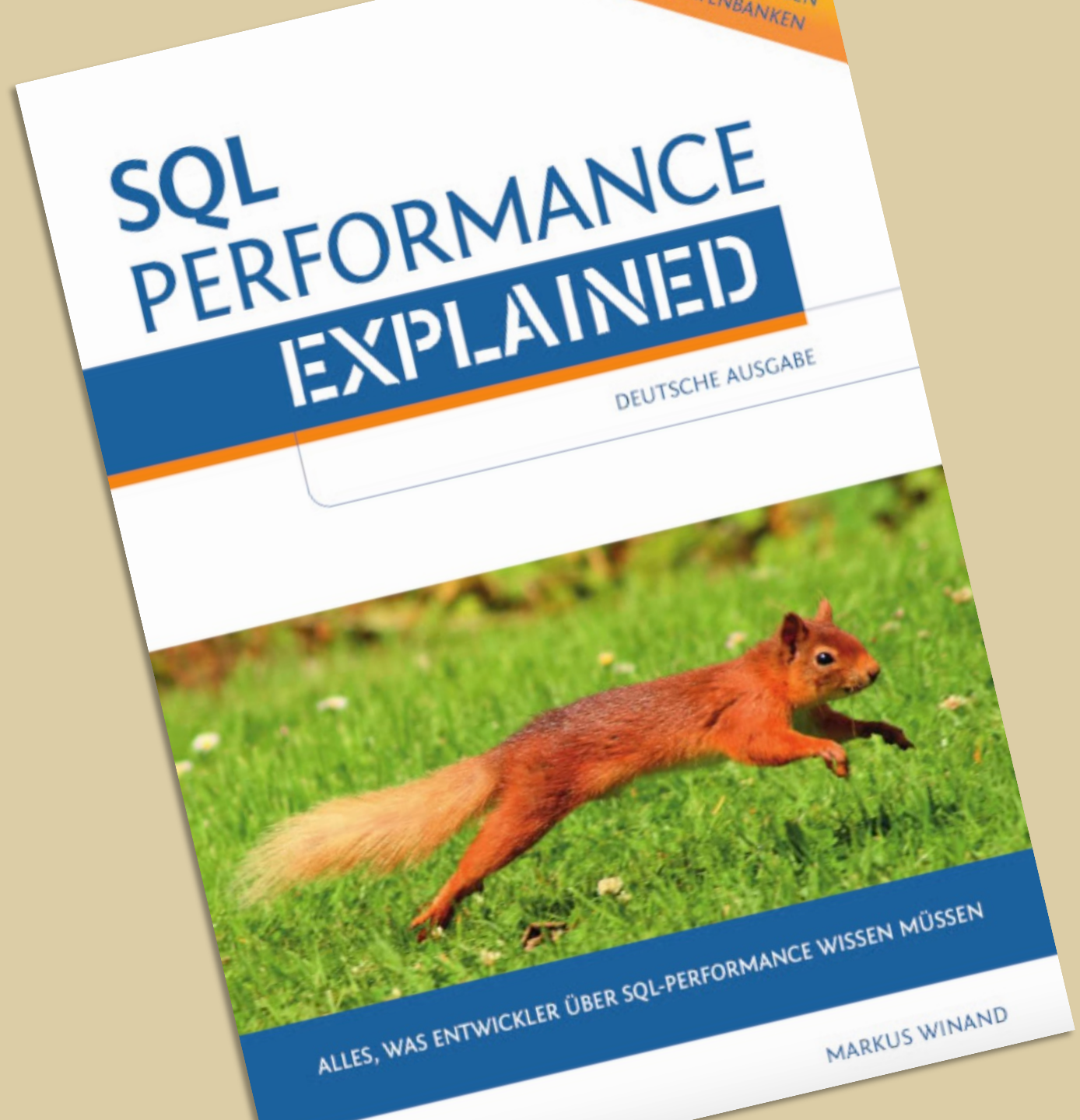


3

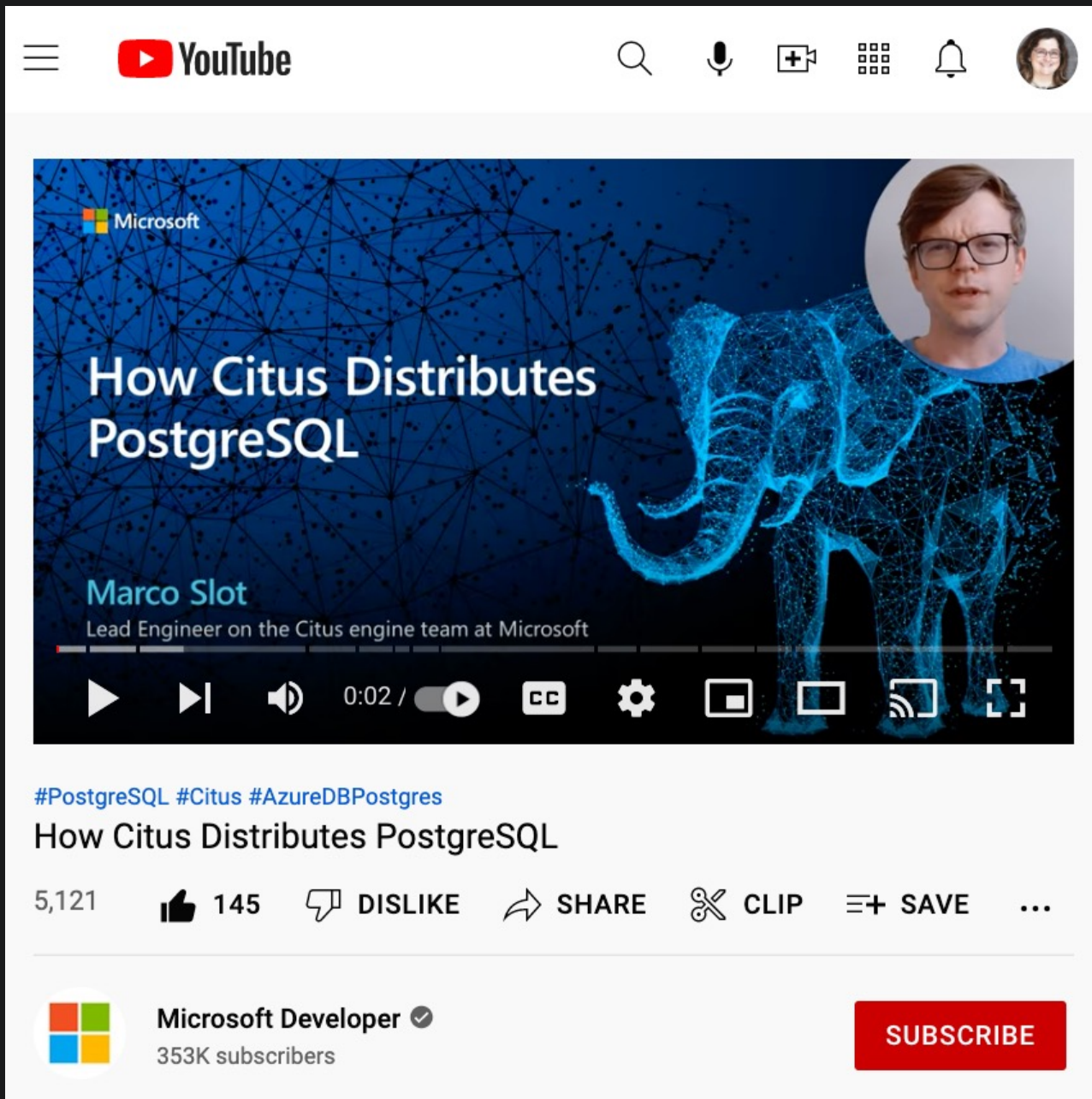
Write a Book

make it easier for newcomers to use Postgres

Source: sql-performance-explained.com



@clairegiordano



The image shows a YouTube video player interface. At the top is the YouTube logo and navigation icons. The video player shows a thumbnail for a video titled "How Citus Distributes PostgreSQL" by Marco Slot, a Lead Engineer on the Citus engine team at Microsoft. The thumbnail features a blue wireframe elephant on a dark blue background with a network pattern. Below the video player, there are hashtags (#PostgreSQL, #Citus, #AzureDBPostgres), the video title, view count (5,121), like/dislike counts (145 likes, 0 dislikes), and share/clip/save options. At the bottom, the Microsoft Developer channel is listed with 353K subscribers and a red "SUBSCRIBE" button.

YouTube

How Citus Distributes PostgreSQL

Marco Slot
Lead Engineer on the Citus engine team at Microsoft

#PostgreSQL #Citus #AzureDBPostgres

How Citus Distributes PostgreSQL

5,121 145 DISLIKE SHARE CLIP SAVE ...

Microsoft Developer 353K subscribers SUBSCRIBE

Use Video to show what Postgres can do

Some people learn better via video

Source: aka.ms/how-citus-distributes-postgres

@clairegiordano

Thumbnail

3 hashtags

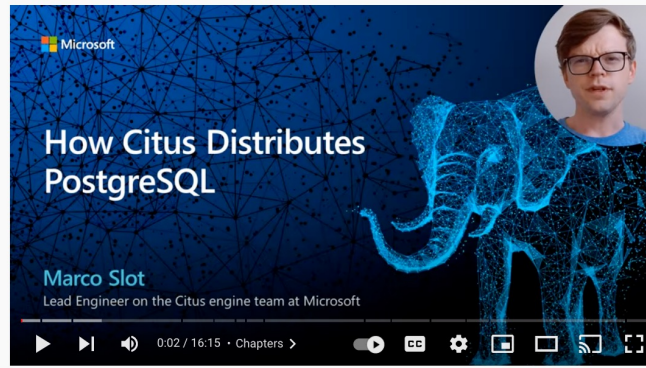
Title

Description

Bookmarks

Links

Chapters



#PostgreSQL #Citus #AzureDBPostgres

How Citus Distributes PostgreSQL

5,121 views · Jul 28, 2021

145

DISLIKE

SHARE

CLIP

SAVE

...



Microsoft Developer
353K subscribers

SUBSCRIBE

This talk by the Citus technical lead is about what Citus is—and how Citus uses the PostgreSQL extension APIs to transform Postgres into a distributed database. Marco Slot explains how Citus works, what you need to know about the new table types Citus introduces (distributed tables and reference tables), what the distributed query planner does, and how Citus handles distributed database transactions.

Marco also explores the 4 application workloads that often need to scale Postgres—including multi-tenant SaaS apps, customer-facing analytics dashboards, as well as more traditional key value workloads with lots of JSON data. Finally, Marco shares the results of performance benchmarks such as HammerDB TPROC-C and YCSB to compare the performance of a single Postgres node, a single Citus node, a Citus 4-node database cluster, and an 8-node cluster.

Citus is available as open source and in the cloud as Hyperscale (Citus), an option in the Azure Database for PostgreSQL managed service. This talk was inspired by a paper Marco delivered on distributed Postgres and Citus at the SIGMOD 2021 conference—link to the paper is below. We hope you enjoy it.

Video bookmarks:

- ▶ 0:30 What is Citus?
- ▶ 1:17 Four workload patterns that need to scale PostgreSQL
- ▶ 5:29 What is a Postgres extension?
- ▶ 5:42 How Citus works
- ▶ 14:28 Performance benchmarking of Citus

Let's connect:

Twitter – @marcoslot, <https://twitter.com/marcoslot>

Twitter – @citusdata, <https://twitter.com/citusdata>

Twitter – @AzureDBPostgres, <https://twitter.com/AzureDBPostgres>



Subscribe to our Citus technical newsletter:
<https://aka.ms/citus-newsletter>



Learn more:

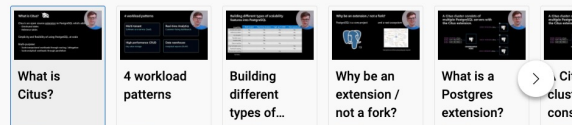
Citus open source repository: <https://github.com/citusdata/citus>

Getting started with Citus: <https://aka.ms/get-started-with-citus>

Citus paper at SIGMOD 2021: <https://aka.ms/citus-sigmod-paper-2021>

#PostgreSQL #Citus #AzureDBPostgres

Chapters



4

Good video metadata
will help your video get
discovered

don't forget to use meaningful
keywords in the filename!

5

Outreach to your other dev communities



The image is a screenshot of a presentation slide from DjangoCon 2019. On the left, a woman, Louise Grandjonc, is standing at a podium, gesturing with her hands. The background behind her has "DIEGO" and "OF" visible. The slide content on the right is titled "Postgres index types" with the subtitle "(And where to find them)". Below the title, it says "Louise Grandjonc" and "DjangoCon 2019". There is a cartoon drawing of a crocodile with its mouth open, showing sharp teeth, and the word "wow" written above it. At the bottom left of the slide, the text "2019 djangocon.us SAN DIEGO" is displayed. At the bottom right, there is a "Full scr" button and a Twitter handle "@louisemeta".

Postgres index types
(And where to find them)
Louise Grandjonc
DjangoCon 2019

wow

2019
djangocon.us
SAN DIEGO

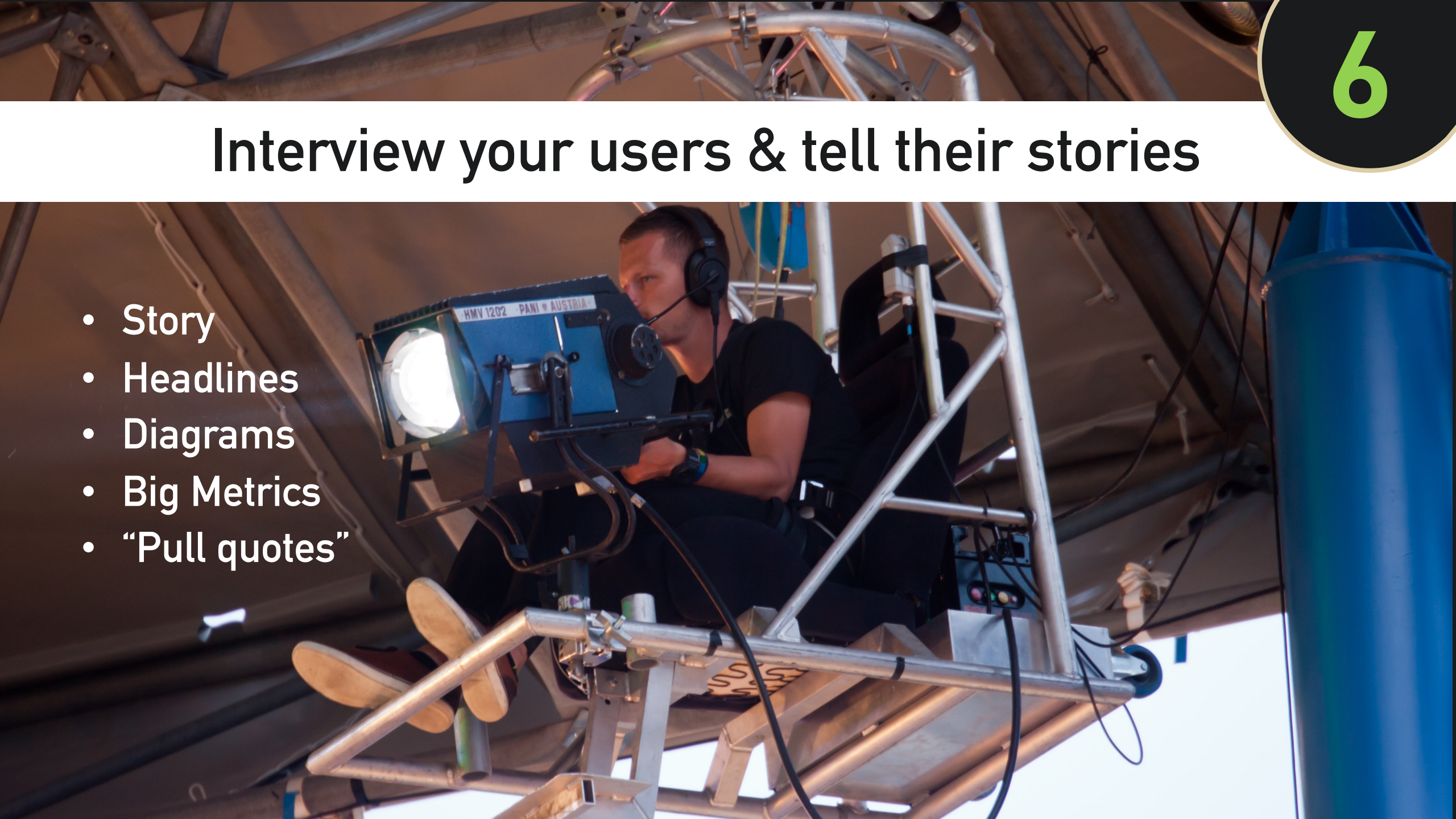
@louisemeta

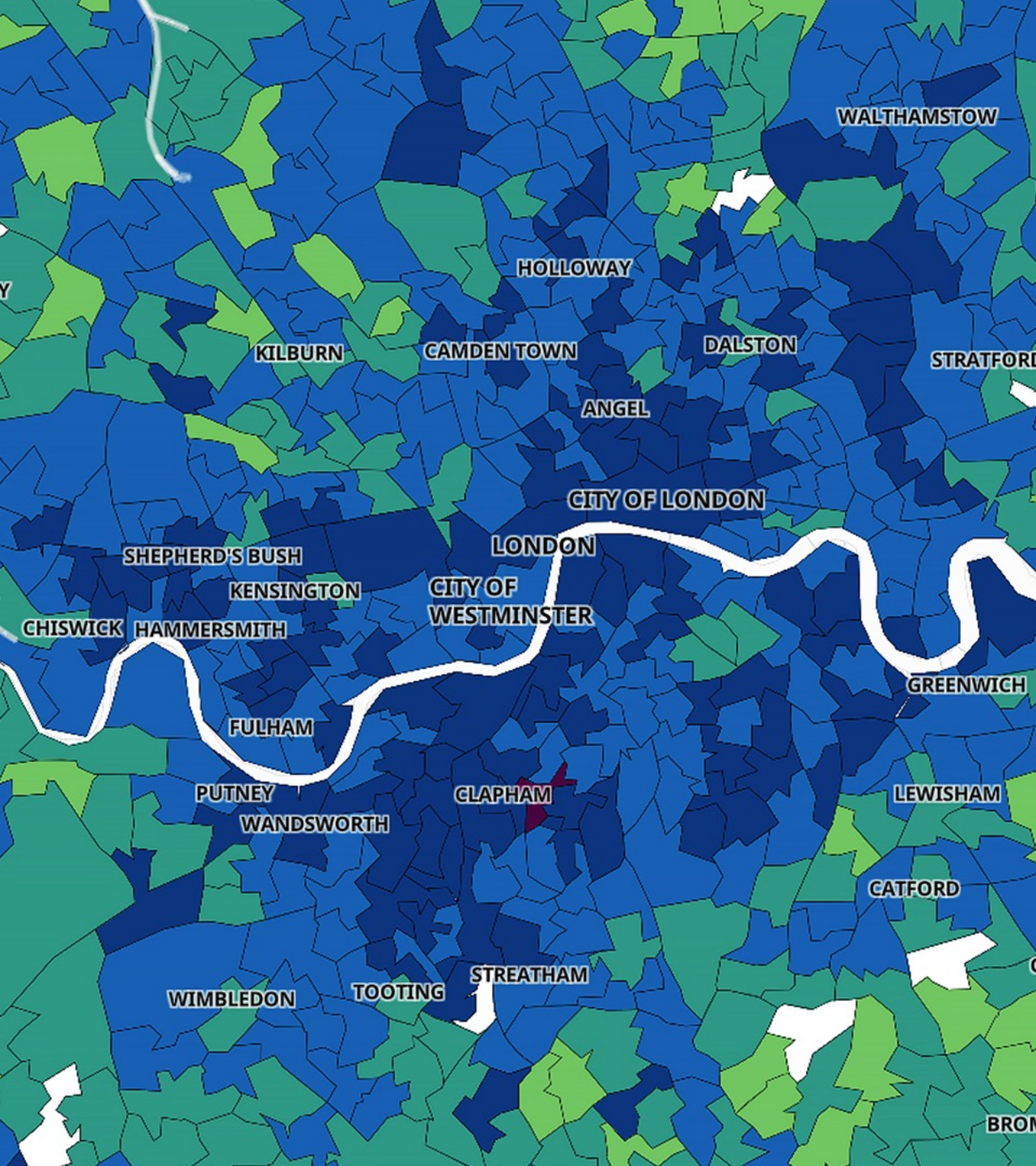
Full scr

Source: Louise Grandjonc talk at DjangoCon US YouTube, DjangoCon 2019

Interview your users & tell their stories

- Story
- Headlines
- Diagrams
- Big Metrics
- “Pull quotes”





Example: Co-authored post w/ Pouria Hadjibagheri of UKHSA on implementation of UK COVID-19 dashboard

- ~77.9K views
- aka.ms/blog-ukcovid19-dashboard-citus

7

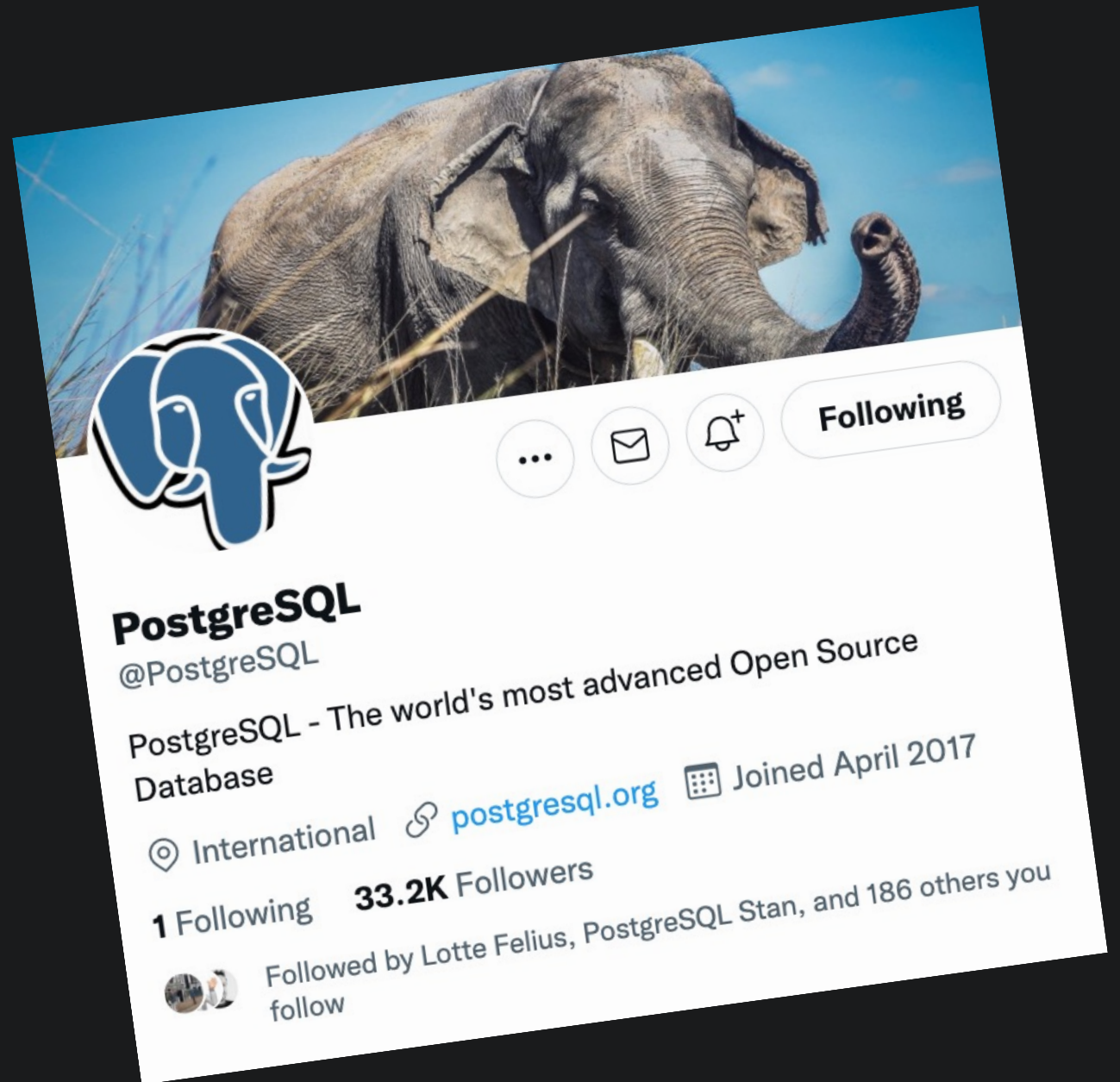
If you build it, they will NOT come. PROMOTE.



7

Promote on Twitter

learnings,
ideas,
talks,
meetups,
TIL, QOTD, OH,
how-to



7

Promote on Twitter

new events, too!



Run, host, & support local meetups



Source: @PostgreSQLNL, @OptimaDataBV

Organize a Postgres event

CITUS CON:
AN EVENT FOR
POSTGRES
2022



38
Sessions

- 20 On-demand talks
- 18 Livestream talks
- 15 Postgres deep-dives

Shine a bright light on
other people's work.
Recognition matters.

- Privately. Publicly. Both.
- Beyond code
- Not just developers

PostgreSQL Person
of the Week

#PostgresFriends

created by Andreas Scherbaum

Report bugs with Postgres

“When you find a bug with Postgres we want to hear about it.”

“Your bug reports play an important part in making PostgreSQL more reliable.”

5. Bug Reporting Guidelines

5.1. Identifying Bugs

5.2. What to Report

5.3. Where to Report Bugs

When you find a bug in PostgreSQL we want to hear about it. Your bug reports play an important part in making PostgreSQL more reliable because even the utmost care cannot guarantee that every part of PostgreSQL will work on every platform under every circumstance.

The following suggestions are intended to assist you in forming bug reports that can be handled in an effective fashion. No one is required to follow them but doing so tends to be to everyone's advantage.

We cannot promise to fix every bug right away. If the bug is obvious, critical, or affects a lot of users, chances are good that someone will look into it. It could also happen that we tell you to update to a newer version to see if the bug happens there. Or we might decide that the bug cannot be fixed before some major rewrite we might be planning is done. Or perhaps it is simply too hard and there are more important things on the agenda. If you need help immediately, consider obtaining a commercial support contract.

Source: [postgresql.org](https://www.postgresql.org)

WHERE to report bugs

“ When you find a bug with Postgres we want to hear about it.”

“ Your bug reports play an important part in making PostgreSQL more reliable.”

Source: postgresql.org

5.3. Where to Report Bugs

In general, send bug reports to the bug report mailing list at `<pgsql-bugs@lists.postgresql.org>`. You are requested to use a descriptive subject for your email message, perhaps parts of the error message.

Another method is to fill in the bug report web-form available at the project's **web site**. Entering a bug report this way causes it to be mailed to the `<pgsql-bugs@lists.postgresql.org>` mailing list.

If your bug report has security implications and you'd prefer that it not become immediately visible in public archives, don't send it to `pgsql-bugs`. Security issues can be reported privately to `<security@postgresql.org>`.

Imaginary email....

To: **pgsql-hackers**
Lists: **pgsql-hackers**

Just tested our
application FOOBAR with
the beta release
candidate for Postgres
15 and wanted to let y'all
that it worked!

12

User testing

your own applications!

new features

recent docs updates

---tell the PG community when
things work

Help with message translations

Mailing list: [pgsql-translators](mailto:pgsql-translators@postgresql.org)

More info: babel.postgresql.org

Instructions in Postgres docs:
“Native Language support”

NLS Status Tables

The tables below show which areas of the PostgreSQL source are prepared for translation, and the progress of those translations. The numbers are percents of translated messages. By following the links, you can download freshly baked PO files that are merged up against the latest program sources. No PostgreSQL source tree is needed for translation work.

The **highlighted** boxes are the files that have a sufficient fraction of strings translated to be considered for release. Files that are **100%** translated are specially highlighted. All other files will not be part of the PostgreSQL release.

The grey numbers are translations that do not exist yet, but where this web site has initialized the translations files for you with data from the existing translations for the same language.

You can work in several branches. Please see the status information below and follow the announcements on the [mailing list](#) ([subscribe](#)) about when a branch is about to be released.

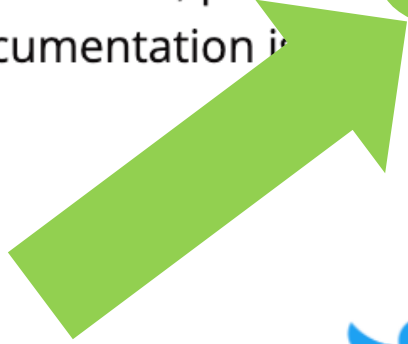
Master

Last update: Tue Apr 12 2022 14:51:48 GMT-0700 (Pacific Daylight Time)

	af	cs	de	el	es	fa	fr	he	hr	hu	id	it	ja	ko	nb	nl
ecpg (131 strings)		93	97	97	97		100	3	3	1	10	90	97	93	2	1
ecpglib (32 strings)		100	100	100	100		100					100	100	100		
initdb (186 strings)		80	82	82	82		100	38	5	1	17	39	82	80	5	6
libpq (263 strings)	19	84	95	93	95		100	63	17	1	2	72	93	84	17	
pg_amcheck (93 strings)		21	94	94	94	1	100	12		1	2	13	94	21	5	

Submit correction

If you see anything in the documentation that is not correct, does not match your experience with the particular feature or requires further clarification, please use **this form** to report a documentation issue.



[Privacy Policy](#) | [Code of Conduct](#) | [About PostgreSQL](#) | [Contact](#)

Copyright © 1996-2022 The PostgreSQL Global Development Group

Source: postgresql.org

14

Submit corrections to documentation

Clarify

Disambiguate

Help new people understand

@clairegiordano

Submit documentation comment

Found something in the documentation that is incorrect, does not match your experience with a particular feature, or requires further clarification?

Please fill out the form below with your name, email, subject, and a detailed description about what you are commenting on. After clicking the button below, an email will be sent to the pgsql-docs@lists.postgresql.org mailing list.

By submitting this form, you agree that all of its contents, including your personal information as listed, will be posted to the public pgsql-docs@lists.postgresql.org mailing list, and archived in the public list archives.

Your Name:

Your Email:

Submit corrections to documentation

Clarify

Disambiguate

Help new people understand

15

Stickers are a thing

Promote your PG projects

Spark unexpected conversations



Use your design skills

Graphics & visuals pull people in

- Citus elicorn
- Citus Con pink elephant
- Postgres Activity Book

Source: Citus “sharded” elicorn mascot

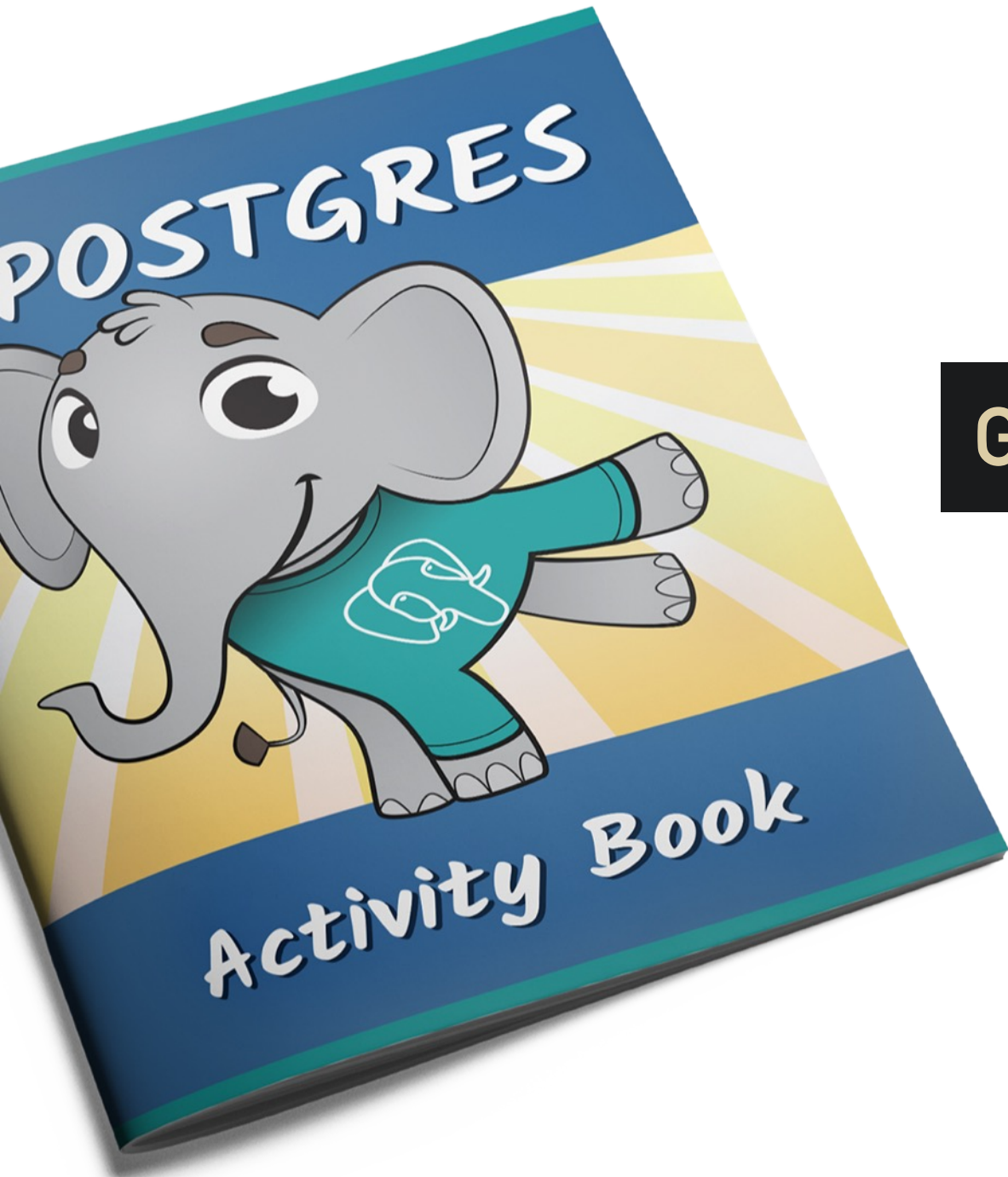


Use your design skills

Graphics & visuals pull people in

- Citus elicorn
- Citus Con pink elephant
- Postgres Activity Book





Use your design skills

Graphics & visuals pull people in

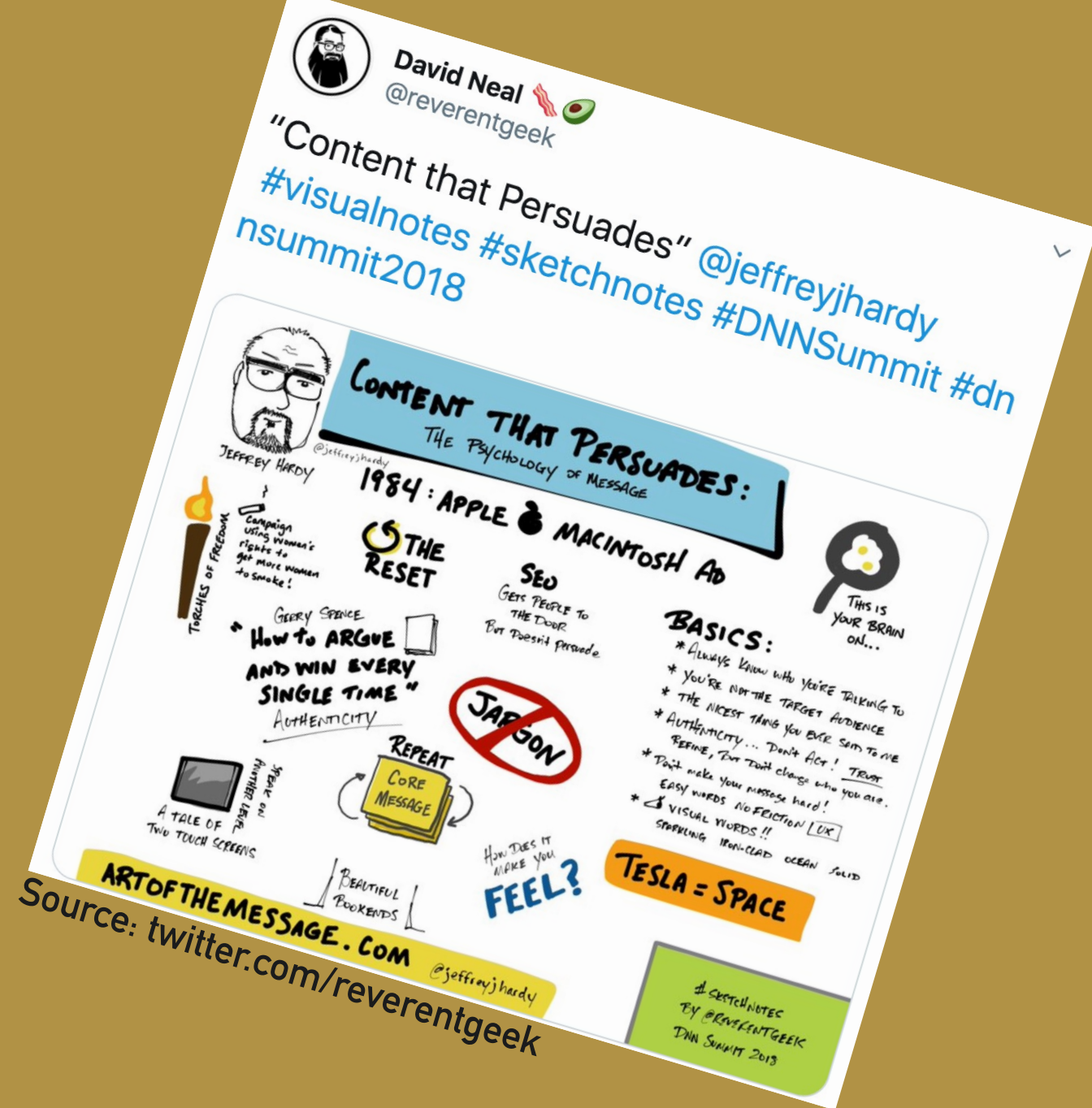
- Citus elicorn
- Citus Con pink elephant
- **Postgres Activity Book**

17

Use "Sketchnotes" to share your learnings

Combine words, lettering styles, color, sketches, bullets

@clairegiordano





Make people feel welcome

Answer questions.
Lots of questions.

Slack. IRC. Stack Overflow.
Mailing lists.

18 Ways to Contribute to Postgres—Beyond Code

1. Write
2. Give a Talk
3. Write a book
4. Video
5. Outreach to other dev communities
6. User stories
7. Promote, tweet too
8. Run & support Meetups
9. Organize a PG event
10. Shine light on others
11. Report bugs
12. User testing of stuff
13. Translations
14. Doc feedback
15. Stickers
16. Design skills
17. #sketchnotes
18. Welcome people in, & answer lots of questions

@clairegiordano

ONE MORE FLIPPINGLY AMAZING

FIBONACCI SPIRAL IN NATURE





Thank you

@clairegiordano

@citusdata • @CitusCon • @AzureDBPostgres



CITUS CON:
AN EVENT FOR
POSTGRES
2022

